development, next-generation **Tang-class** SSBNs are expected to have improved stealth and carry JL-3 missiles with a longer range

↘ Shang-class (SSN)

Numbers: **6**

First commissioned in 2006, these nuclear attack submarines have advanced sensors and sonar and carry YJ-18 and YJ-82 anti-ship cruise missiles. Upgraded version has improved stealth. Under development Sui-class SSNs are expected to feature advanced technology with reduced noise levels and better sensors

↘ Song-class (SSK)

Numbers: **13**

These diesel-electric attack submarines were inducted between 1999 and 2006 and carry torpedoes, mines and anti-ship missiles

↘ Yuan-class (SSK)

Numbers: **20+**

The backbone of China's submarine fleet, some variants are equipped with Air-Independent Propulsion (AIP) systems, giving them extended underwater endurance. Newer variants with better stealth and improved capabilities are being introduced. China also has 12 older **Ming-class** diesel-electric submarines





PAKISTAN 3

↘ Khalid-class (SSK)

Numbers: **3**

French-origin Agosta-class submarines commissioned between 1999 and 2006. Crucially, equipped with AIP systems and Exocet anti-ship missiles. Have undergone upgrades

↘ Hangor-class (SSK)

Numbers: **8 (in development)**

To modernise its conventional diesel-electric fleet, Pakistan has turned to China to provide 8 new-generation Yuan-class subs. Expected to feature AIP systems and advanced weapons systems. First deliveries expected in 2024; all vessels planned to be operational by 2028

proved Project Delta, under which India would lease six nuclear-powered SSNs (which lack the capability to launch long-range ballistic missiles) from Russia. However, the desired Akula-class SSNs were delayed due to Russia's preoccupation with the Ukraine conflict, the navy has now moved a proposal to indigenously build two SSNs.

India's efforts to modernise its conventional submarine fleet also led to the 2005 contract for six French Scorpene-class diesel-electric attack submarines. They were developed in collaboration with the French company Naval Group and built by MDL. The first of these submarines, the INS *Kalvari* (named after India's first submarine, and which lends its name to the class), was commissioned in 2017. Four more Kalvari-class submarines were inducted in the following years, with the sixth one expected later this year. Sensing the growing need for advanced conventional submarines, the Navy in 2021 approved the acquisition of three additional Kalvari-class submarines. MDL is in advanced talks for a potential Rs 35,000 crore deal for the three vessels that will be larger and more advanced than the first six.

With advanced stealth features and the capability to launch Exocet anti-ship missiles, the Kalvari-class submarines are a significant upgrade over the Sindhughosh and Shishumar classes, but they lack the AIP technology and sensors of the most advanced conventional subs. The Project 75(I) submarines would address this shortcoming.

"The contract for Scorpenes was signed in 2005, but it ran behind schedule and cost almost twice as much. This led to the obsolescence of equipment," says a serving navy submariner. According to him, the lack of AIP is a major limitation in deploying subs in anti-submarine warfare. "Even Pakistan," he says, "has had AIP capability on its submarines since 2005, ironically delivered by the French."

FULL TRANSFER OF TECHNOLOGY

Despite its strategic importance, P75(I), too, has faced delays, with momentum picking up in 2021. Vice Admiral Singh believes that India desperately needs P75(I), but he is apprehensive about its projected timeline (seven years for the first sub). He points out that both Indian submarine manufacturing shipyards—the MDL in Mumbai and the L&T shipyard near Ennore in Tamil Nadu—have their hands full. "MDL is gearing up for three additional submarines, while L&T is on a nuclear sub. Unless we find a third shipyard, we do not see Project 75(I) translated into reality," he says.

As for its current status, after the FET by the Indian Navy, the Spanish firm Navantia is at a pronounced disadvantage as they are yet to have a proven AIP. However, being state-owned, they offer crucial governmental support and guarantees. Navantia has offered a submarine based on its new S80 class. In contrast, the German TKMS has built over 170 submarines for 15 navies, and 52 of them have time-tested fuel cell AIP systems that are being offered to India.

Khalil Rahman, chief executive officer, TKMS India, says the proposed deal has the full backing of the German government. He says TKMS promises a full transfer of technology as India has demanded, stressing that Germany will guarantee it. "The design is a derivative of the HDW Class 214. It is adapted to fulfil the P-75(I) requirements. ThyssenKrupp Marine Systems is a forerunner in stealth technology and the only provider of a sea-proven fuel cell AIP system which dramatically increases the range at which a submarine can remain undetected," Khalil tells INDIA TODAY.

Whether it chooses Navantia or TKMS, India must expedite Project 75(I), addressing the constraints that have contributed to the country's declining fleet size. ■